

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS:

Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly. **Questions:**

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

*I **Bhargavi. N (192411144)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Roshitha :)



Web Technologies Case Studies

1. Online Symposium Portal

Given Code

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

Missing Semantic Elements

- <header>
- <main>
- <section>
- <footer>
- <label>

HTTP Request

```
POST /register HTTP/1.1
name=Student
email=student@example.com
```

HTTP Response Cycle

1. Browser sends POST request
2. Server validates data
3. Server stores registration details
4. Server sends response
5. Browser displays confirmation page

Accessibility Improvements

- Add labels for all inputs.
- Use required attributes.
- Provide placeholders.
- Ensure keyboard accessibility.

2. Cross-Browser Compatibility Issue

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Issues Identified

- Missing </h1> tag.
- Missing </p> tag.
- No semantic elements used.

Correct HTML Structure

```
<header>
  <h1>Welcome</h1>
</header>

<main>
  <section>
    <p>About Us</p>
  </section>
</main>
```

Cross-Browser Compatibility Tips

- Validate HTML using W3C Validator.
- Use CSS resets.
- Test in Chrome, Edge, Firefox, and Safari.
- Use responsive design techniques.

3. Form Submission Failure

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Problems

- Button type is set to "button".
- GET exposes credentials in the URL.
- Passwords should never be transmitted using GET.

Reason for Failure

Since the button type is **button** instead of **submit**, clicking it does not send an HTTP request to the server.

Correct Implementation

```
<form action="/submit" method="POST">
  <input type="text"
    name="username"
    required>
  <input type="password"
    name="password"
    required>
  <button type="submit">
    Login
  </button>
</form>
```

Security Improvements

- Use POST instead of GET.
- Enable HTTPS.
- Hash passwords before storing.
- Implement server-side validation.
- Use secure session management.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand